

Sifted Disks

*reducing the number of sample points
retaining randomness
improving quality*

Mohamed S. Ebeida

Ahmed H. Mahmoud - Muhammad A. Awad - Mohammed A. Mohammed

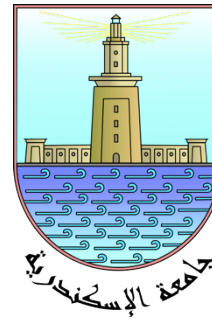
Scott A. Mitchell

7!

Alexander Rand - John D. Owens



**Alexandria
University**



Eurographics 2013

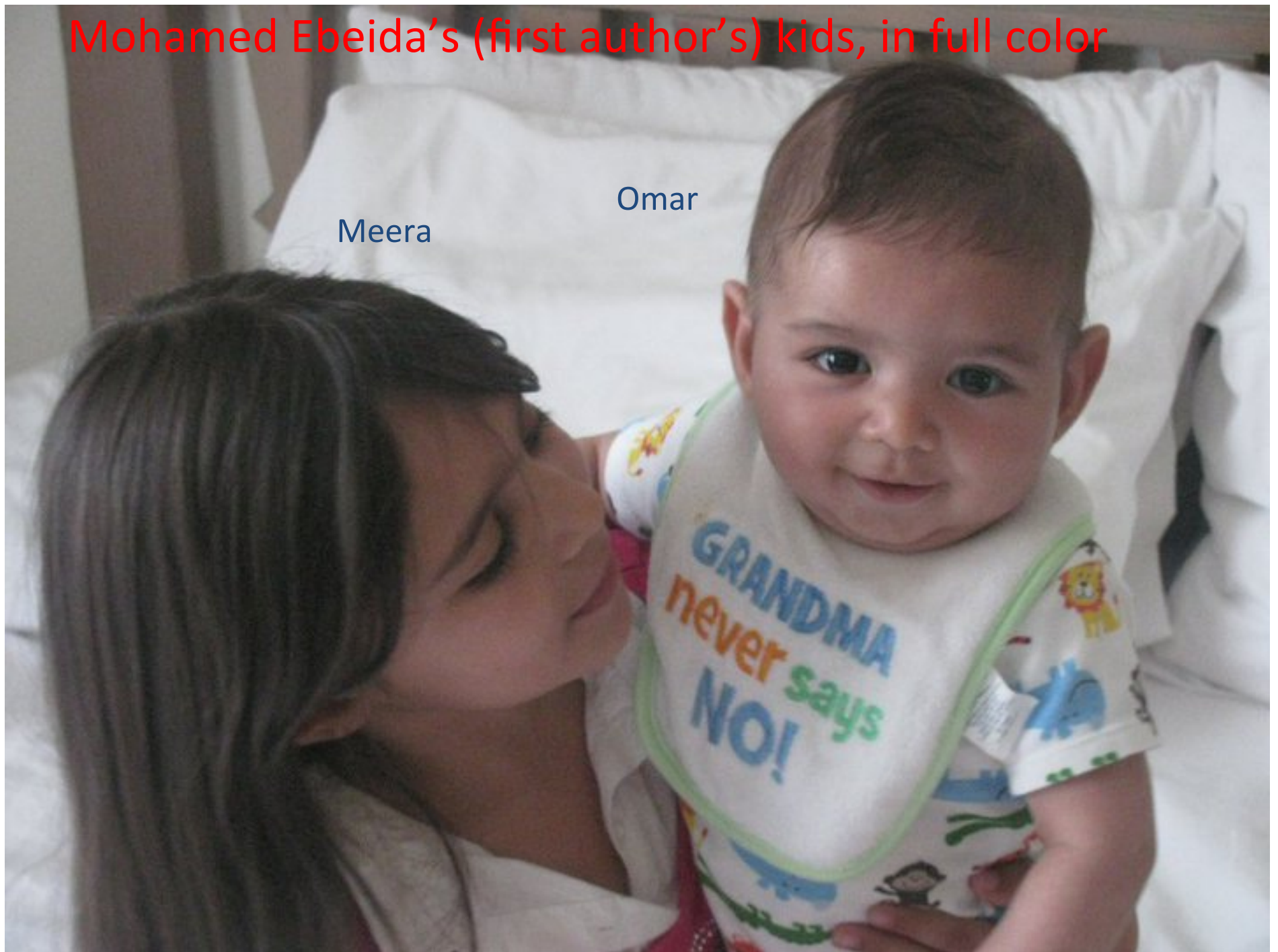
presenter = Scott



Sandia is a multiprogram laboratory operated by Sandia Corporation, a Lockheed Martin Company,
for the United States Department of Energy's National Nuclear Security Administration
under contract DE-AC04-94AL85000.



Mohamed Ebeida's (first author's) kids, in full color



Application

1. grayscale -> sizing function for



2. Stippling via
Maximal
Poisson-Disk
Sampling

3.
Sift points

Replace 2 for 1.

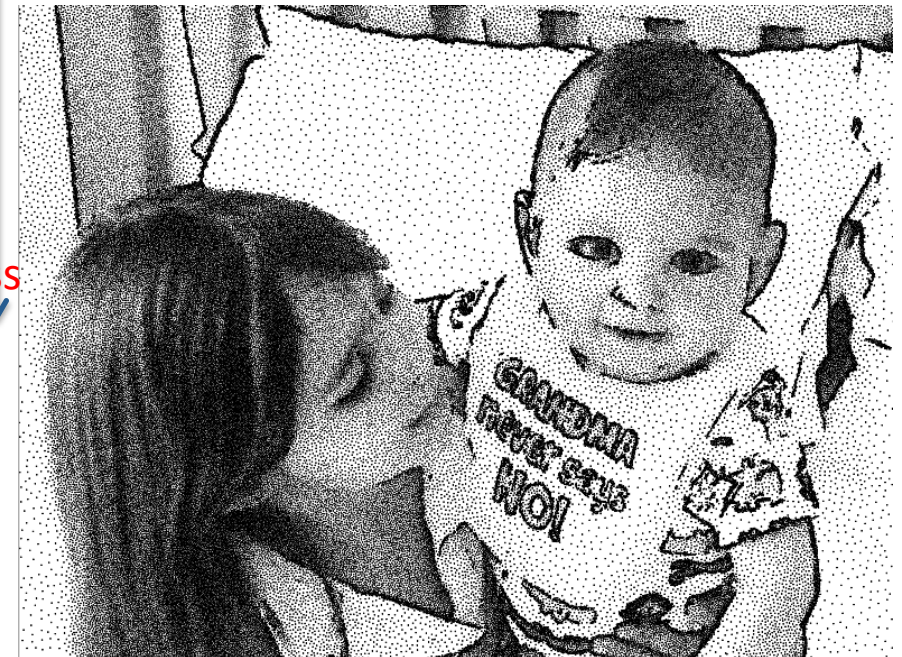
Respect original
sizing function.

Fewer points
Minimal quality loss



edge-detect

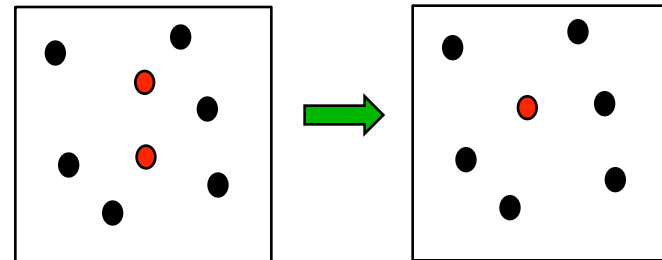
Universally lighter,
but features still
distinct





Overview

- Input: point sample distribution
Poisson disks, Delaunay Refinement
 - Sizing function
 - Adheres approximately
- Observe: **other distributions also respect sizing function, might be smaller**
- Process
 - Replace points 2-for-1
 - Adhere to sizing function
- Result
 - Goal 1: Fewer points --- how many?
 - Goal 2: Retained randomness --- surprise!



Input point sets – improve them

Sifting points from

DR Delaunay Refinement

ODR Delaunay Refinement w/ Off-centers

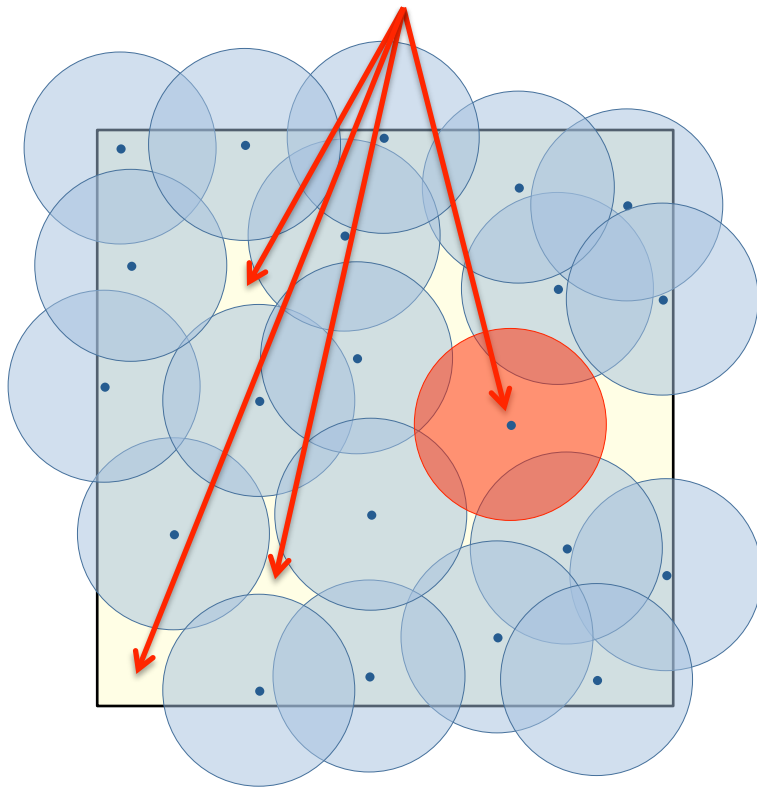
MPS Maximal Poisson-disk Sampling

MPS – Maximal Poisson-disk Sampling

new disk

global uniform random locations

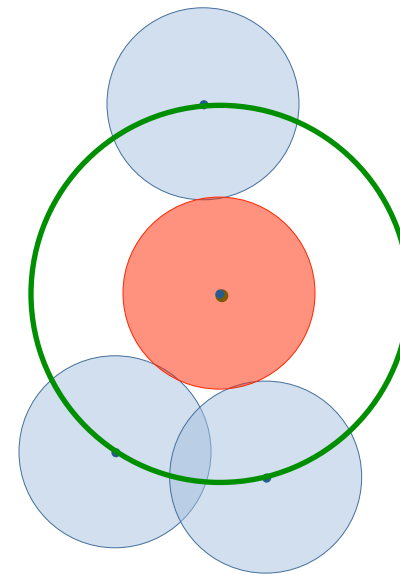
outside prior disks



DR – Delaunay Refinement

new disk

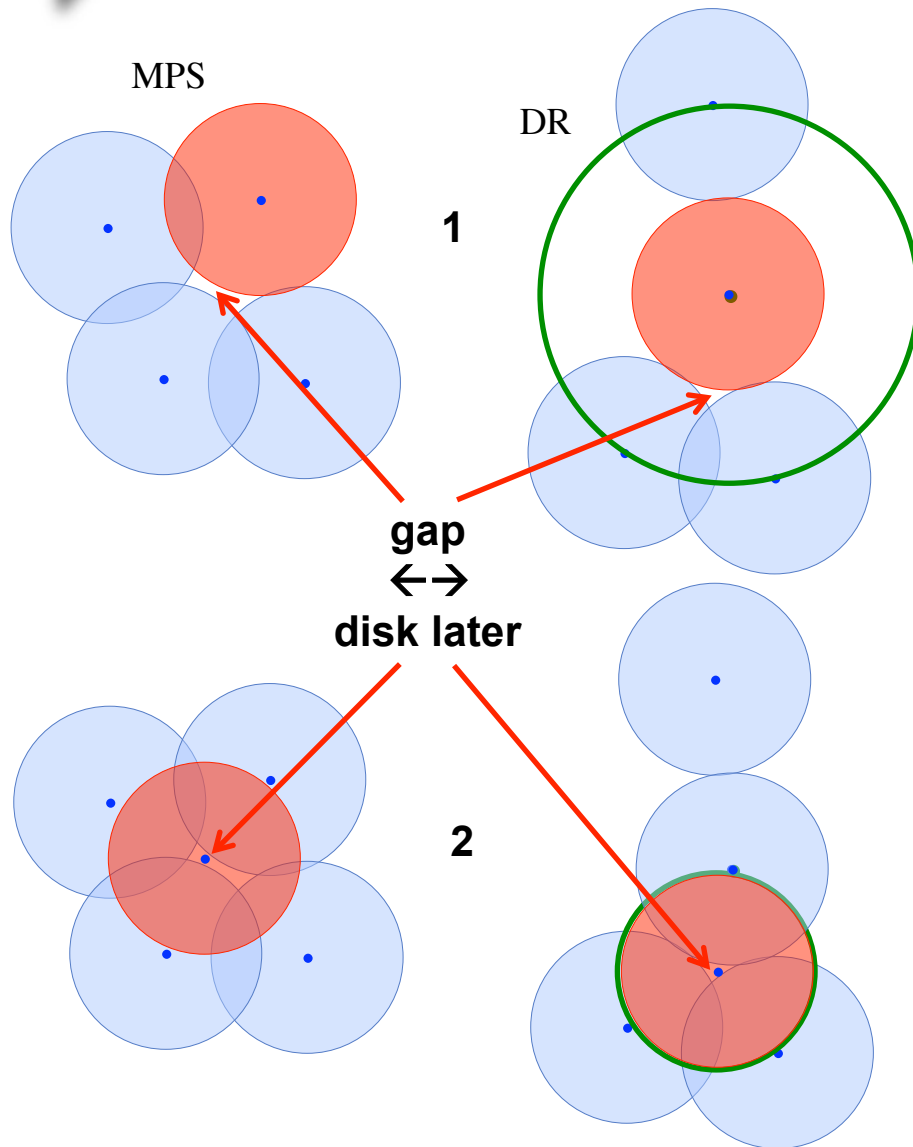
center of large empty dual circle





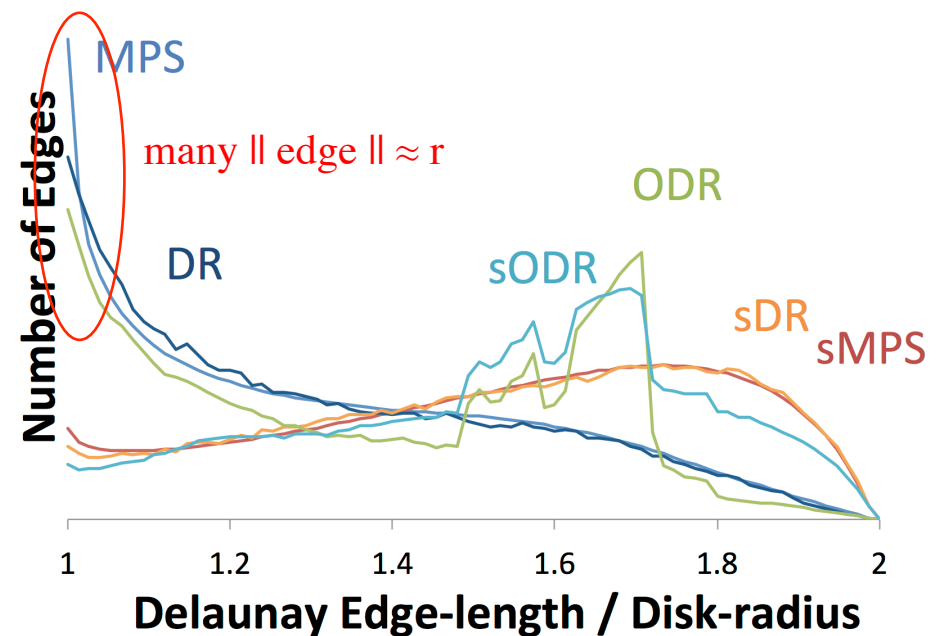
Problem

Painting Yourself Into a Corner



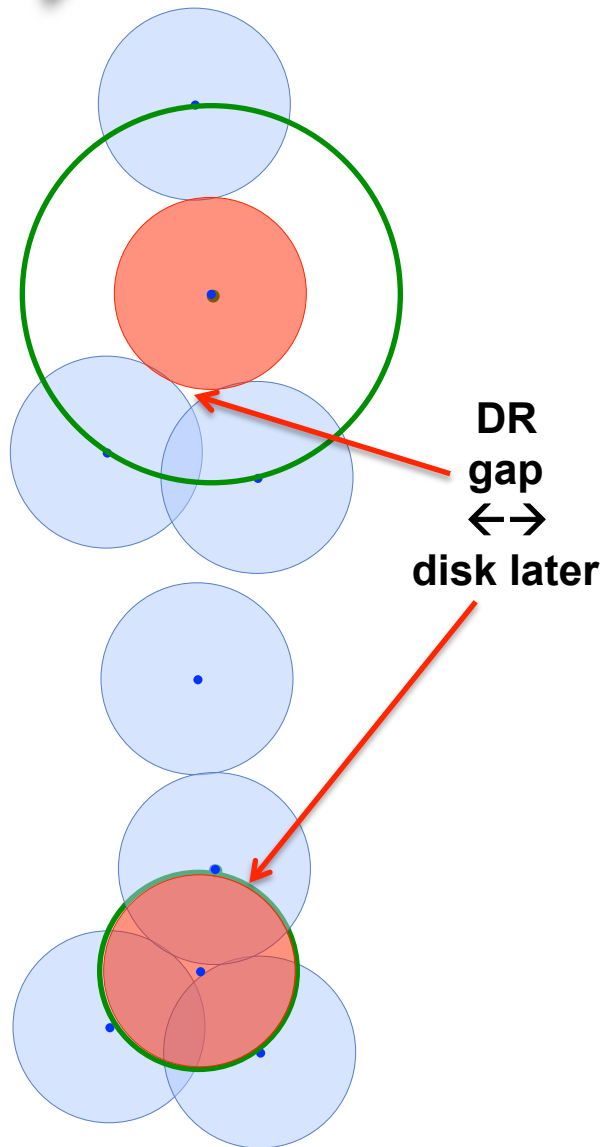
MPS, DR
easy to introduce a small gap
that later forces

- distance = $r + \text{eps}$
- dense sampling



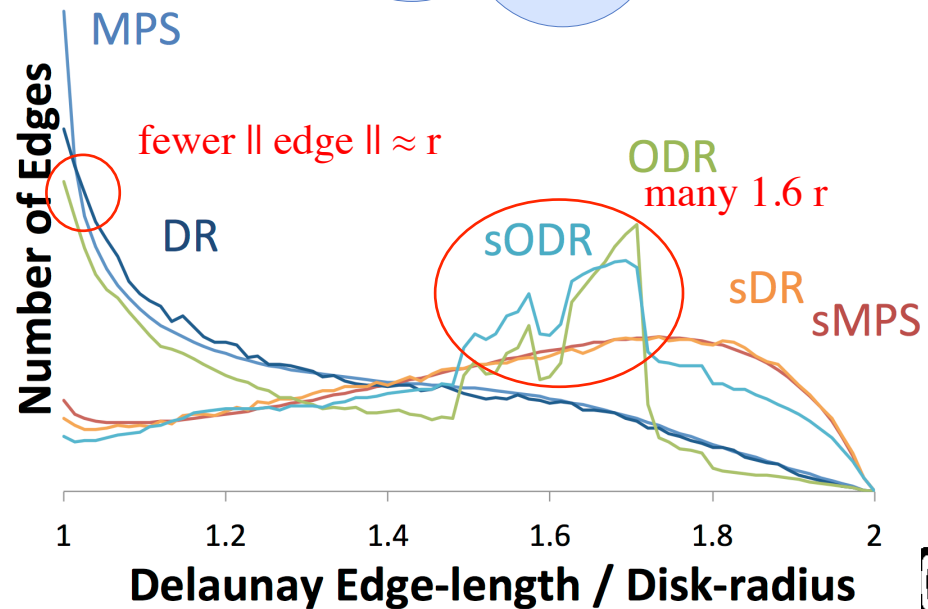
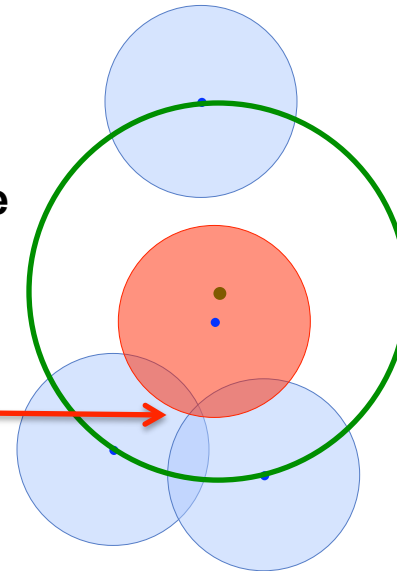


DR Solution off-centers DR (ODR)



move disk
towards short edge

ODR
no gap



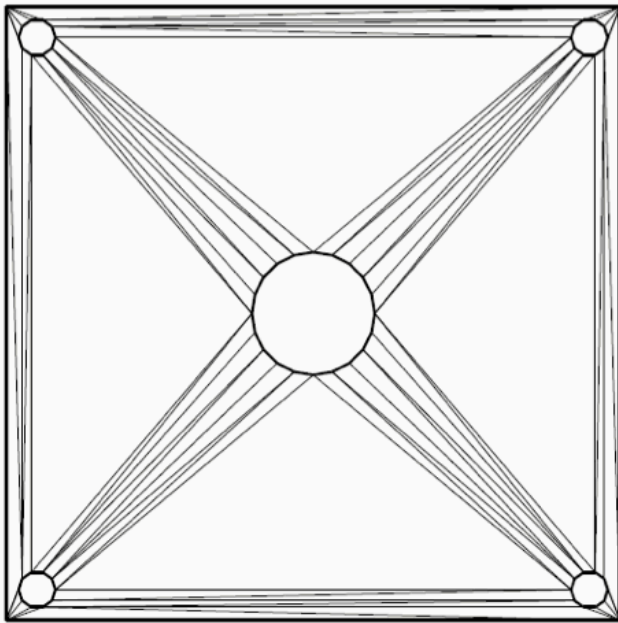
Offcenters reduces density... by a lot for non-uniform sizing functions

But we will focus on
 $r = 1$, uniform

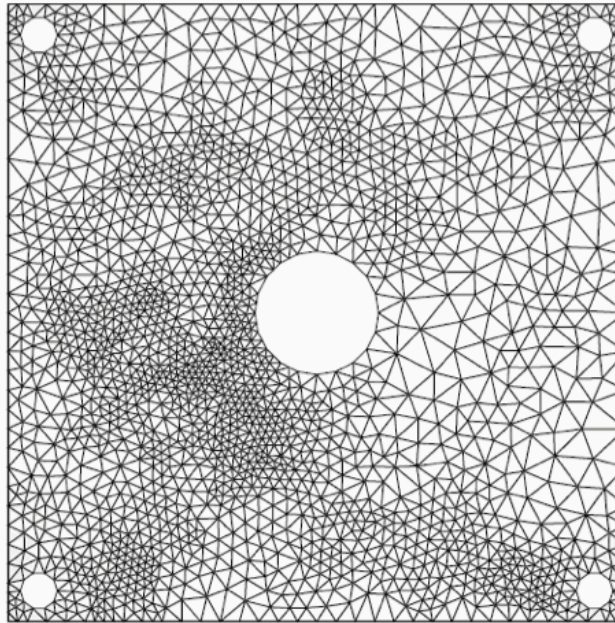
10

Alper Ungör

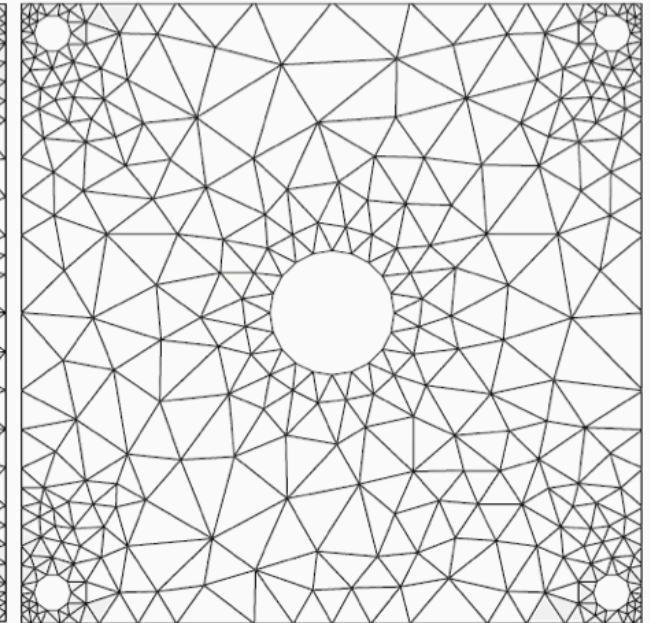
“Off-centers: A new type of Steiner points for computing size-optimal quality-guaranteed Delaunay triangulations”



(a) input



(b) DR



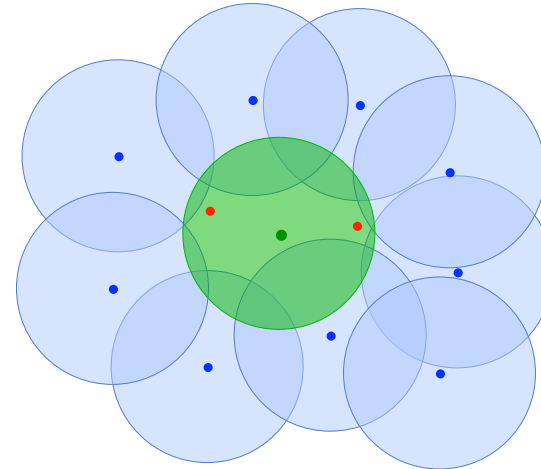
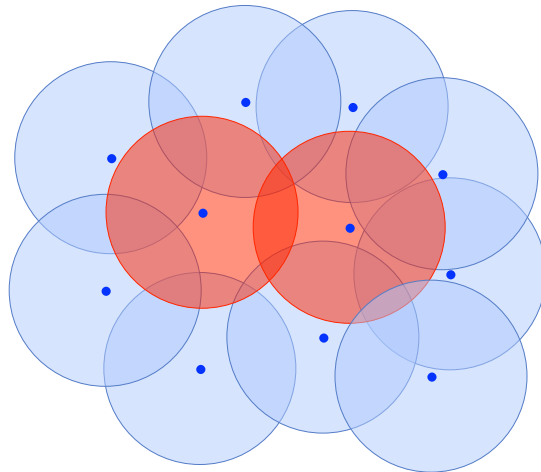
(c) ODR

Fig. 4. Input PSLG is a plate with five holes described by 64 points and 64 segments. Smallest angle in the initial triangulation (a) is about 1° . Smallest angle in both output triangulations is 34° . Circumcenter insertion (`triangle` software) introduces 1984 Steiner points resulting a mesh with 3910 triangles (b). Off-center insertion introduces only 305 Steiner points resulting a mesh with 601 triangles (c).



Our MPS-like Solution: Sifting

- Post-process
- For all pairs of points with overlapping disks
 - Try to replace 2-for-1
 - (Replacing changes the set of overlapping pairs)
- Quit when no pair can be replaced

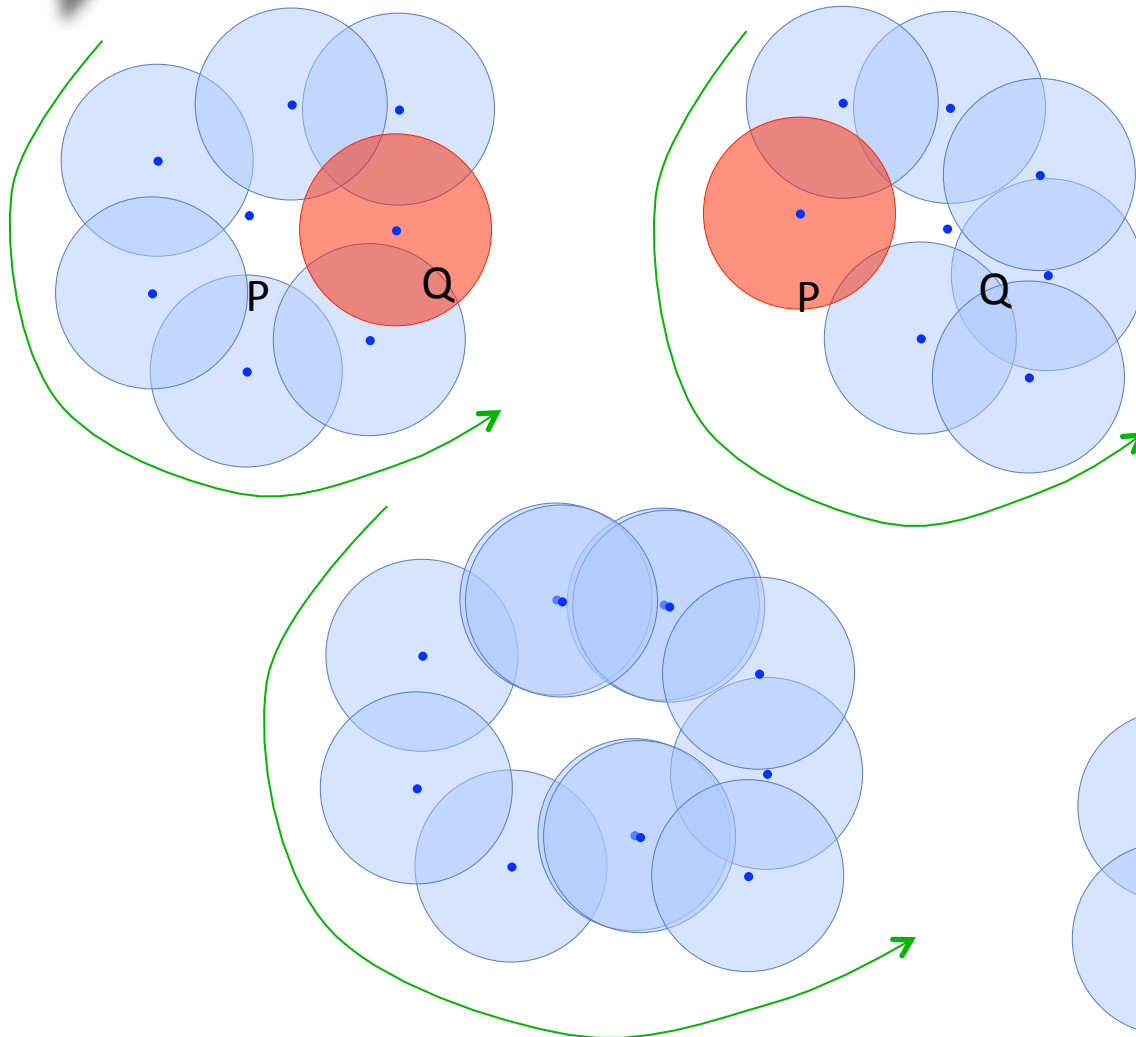


replaced
●



Sifting Algorithm

Gather Boundary Disks



1. Gather disks overlapping P
– (Q)
Sort by angle around P (Q)

2. Stitch lists together
– Replace Q in ListP by ListQ

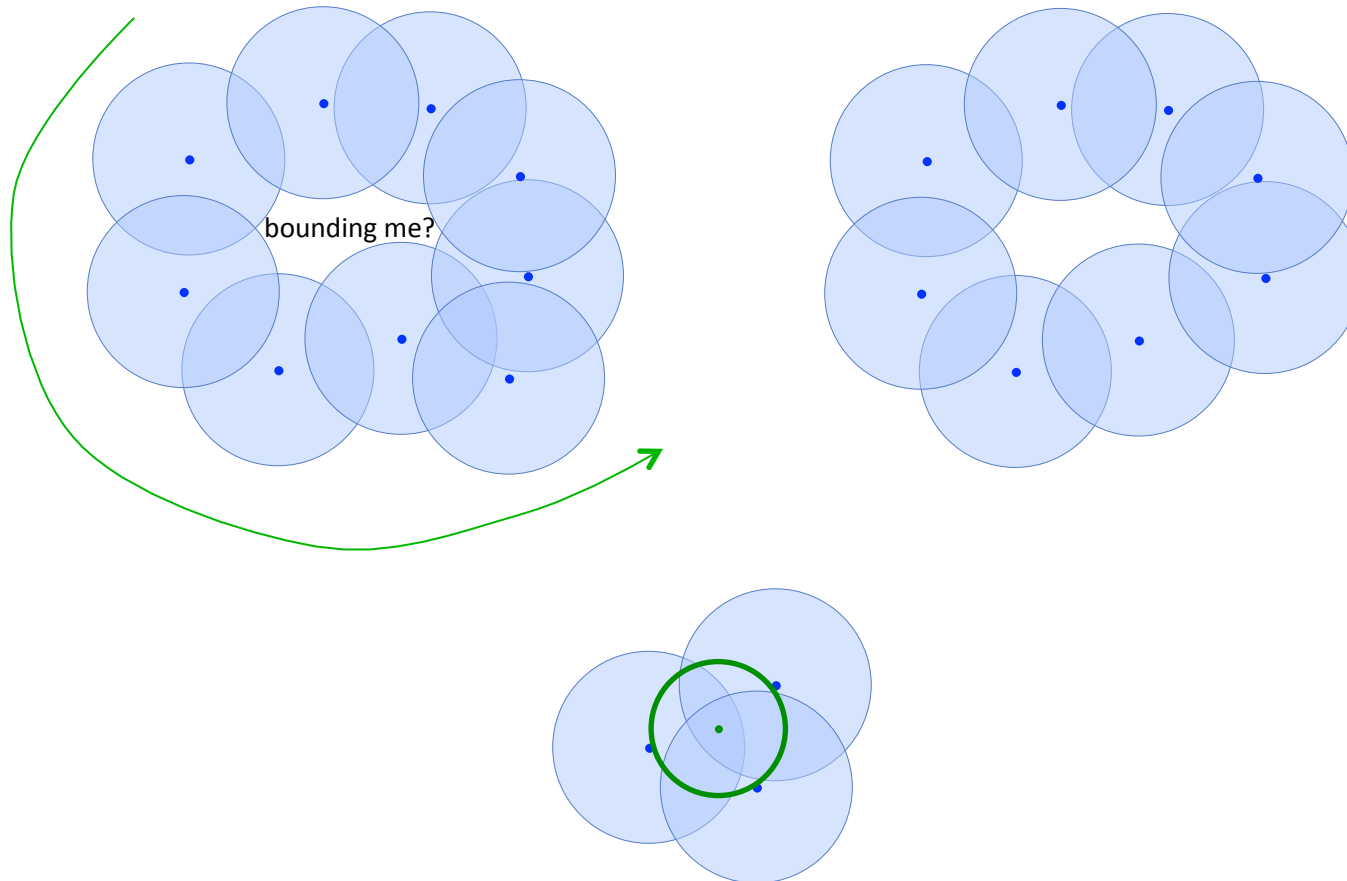
3. Remove duplicate disks



Sifting Algorithm

Winnow Non-Bounding Disks

- Remove disks not bounding the white area
 - Test consecutive disks in list, see if left point of intersection is inside next disk



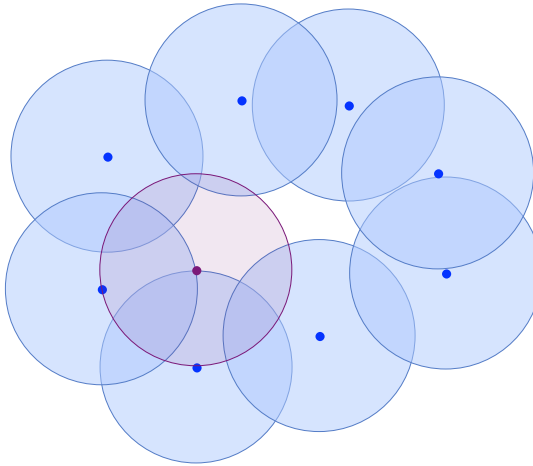
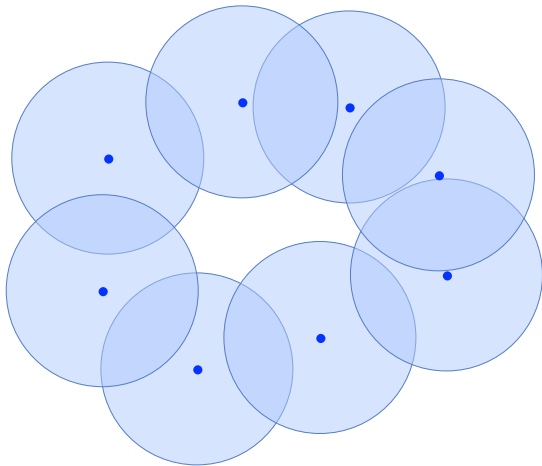
- In-circle test speeds up intersection point test by 3x
technical for non-constant radius, details in paper 😊



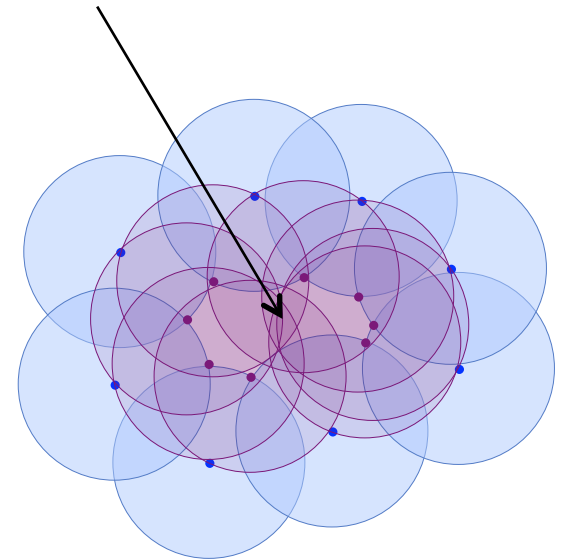
Sifting Algorithm

Exclusion – Inclusion Disks

- A new disk will cover all the white area
 - iff it covers all the corners of intersection
 - Reason: because disks are convex
- Need replacement disk
 - Outside all **sample disks**
 - Inside all **dual corner disks**



Is there a common intersection?



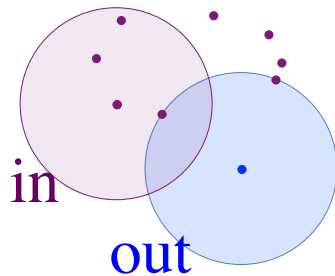


Sifting Algorithm

Search for Random Location – Using “Simple MPS”

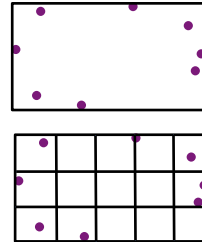
– Problem: find random point that is

- Outside all **sample disks**
- Inside all **dual corner disks**



– Solution

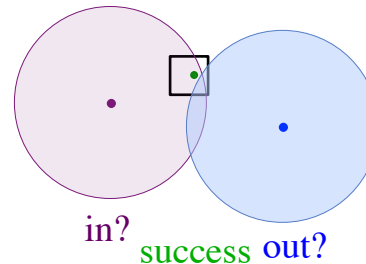
- Simple MPS [Ebeida et al. Eurographics 2012]
- extended for purple inclusion disks
- Flat quadtree
 - approximates acceptable region



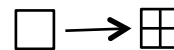
Simple MPS Algorithm Details

Initialize

bounding box of purple corners
subdivide into squares - diagonal about radius

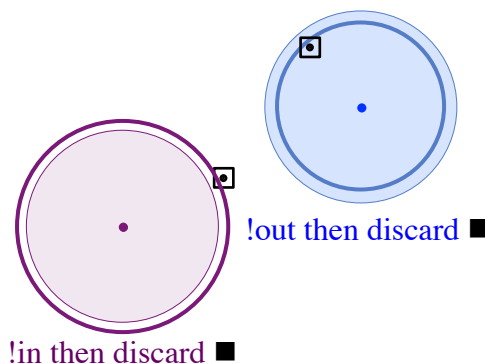


Sample C | #square times |
pick a square
pick a point p in the square
keep p if out-blue & in-purple
success!



Refine all squares

center inside a blue circle - delta? Discard
center outside a purple circle + delta? Discard



Repeat with refined squares

No squares? No replacement exists



How Well Did We Do?

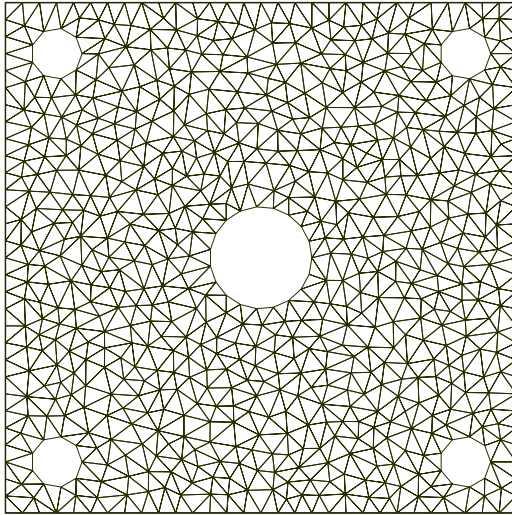
- **Goal 1: Number of points = density**



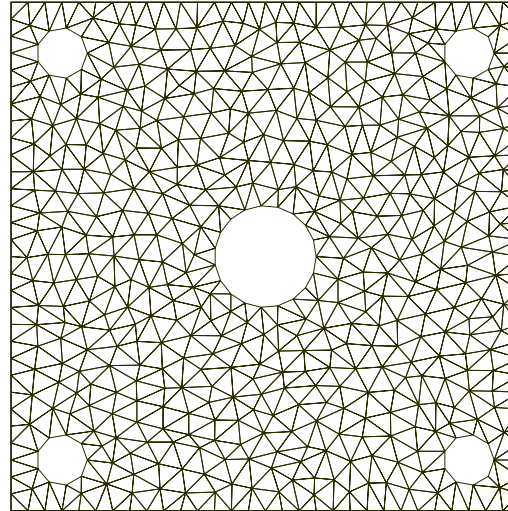
Sifting Improves All Uniform Test Distributions

- Sifting improves, MPS, DR and **further improves ODR**

MPS

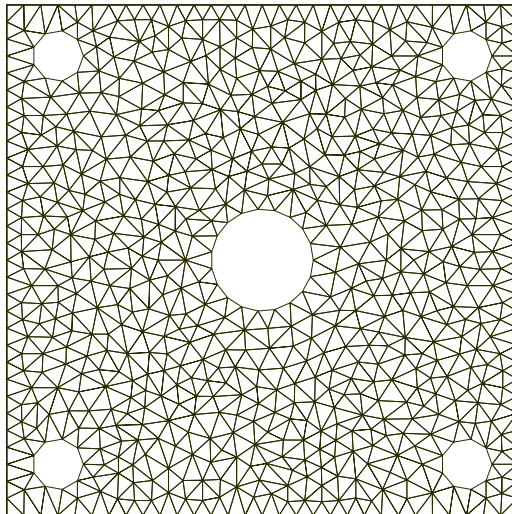


sMPS

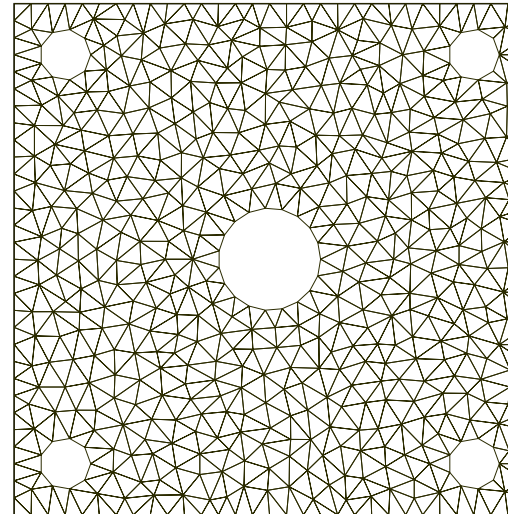


Sift->

DR



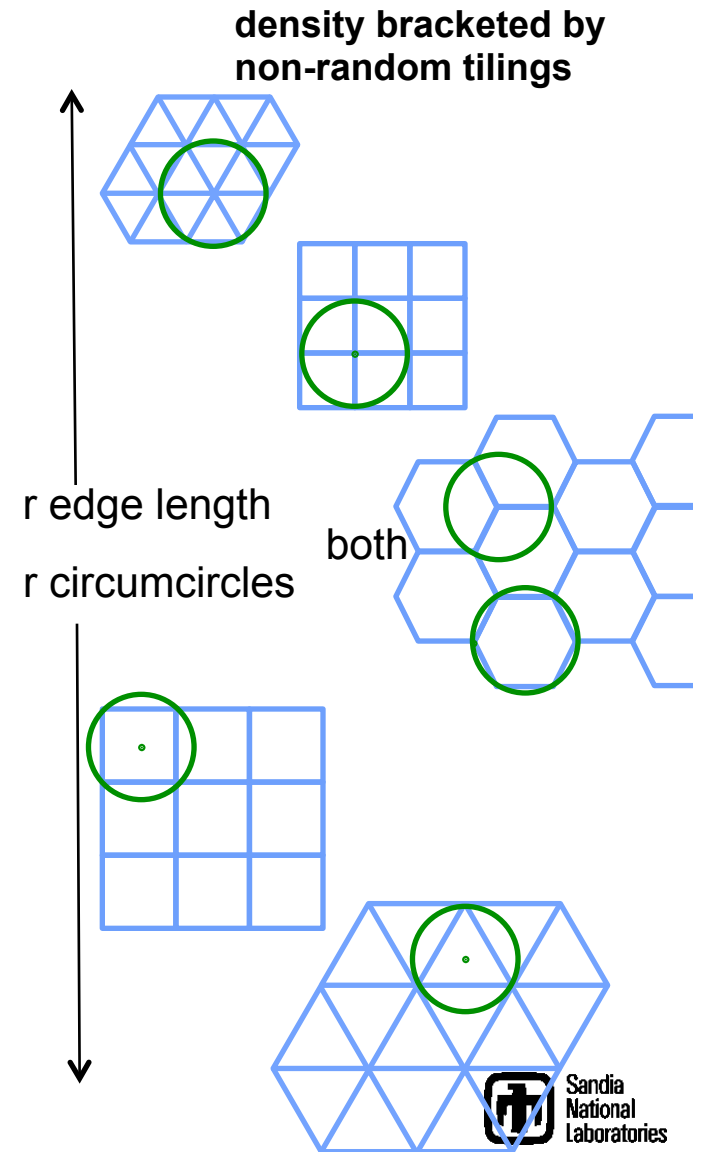
sDR



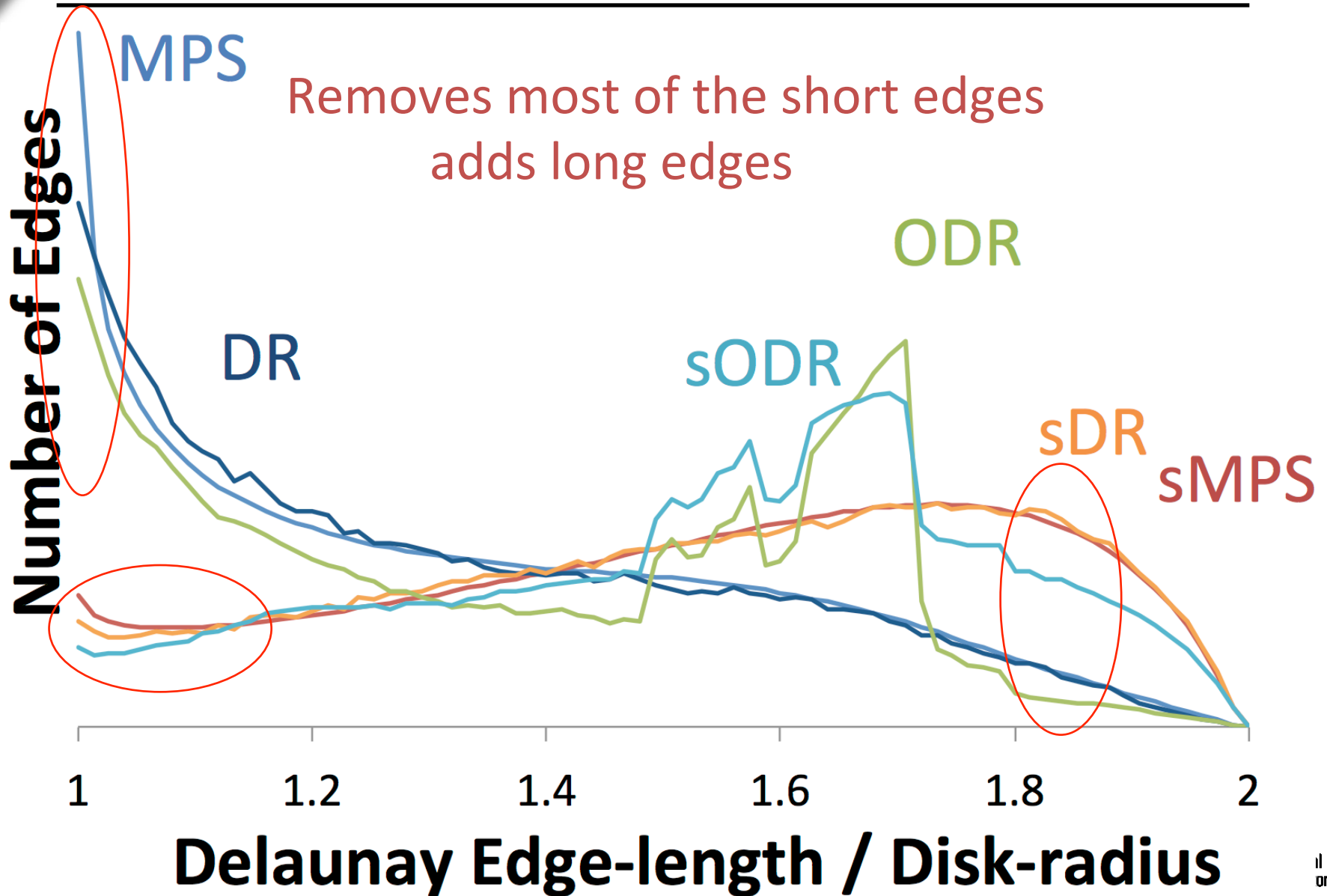


Sifting reduces number of points by $\approx 25\%$

	sample type	point density	relative density	Delaunay edge lengths
densest possible	$\triangle(r)$	$\frac{2}{\sqrt{3}}r^{-2}$	3	$\{r\}$
	$\square(r)$	r^{-2}	2.60	$\{r, \sqrt{2}r\}$
	$\hexagon(r)$	$\frac{4}{3\sqrt{3}}r^{-2}$	2	$\{r, \sqrt{3}r, 2r\}$
input	DR(r)	$0.75r^{-2}$	1.95	$[r, 2r)$
	MPS(r)	$0.70r^{-2}$	1.82	$[r, 2r)$
	ODR(r)	$0.64r^{-2}$	1.66	$[r, 2r)$
	sDR(r)	$0.57r^{-2}$	1.48	$[r, 2r)$
	sMPS(r)	$0.51r^{-2}$	1.33	$[r, 2r)$
sifted	sODR(r)	$0.51r^{-2}$	1.33	$[r, 2r)$
	$\square(\sqrt{2}r)$	$\frac{1}{2}r^{-2}$	1.30	$\{\sqrt{2}r, 2r\}$
sparsest possible	$\triangle(\sqrt{3}r)$	$\frac{2}{3\sqrt{3}}r^{-2}$	1	$\{\sqrt{3}r\}$

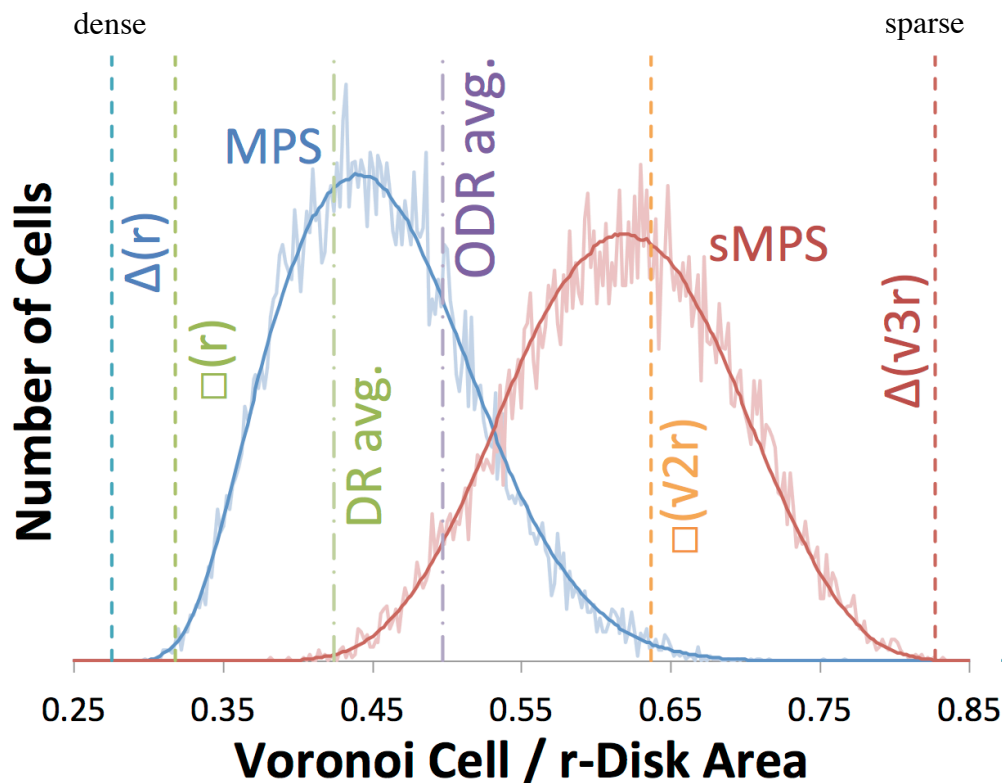


Sifting changes triangulation edge lengths, angles, Voronoi cell squish

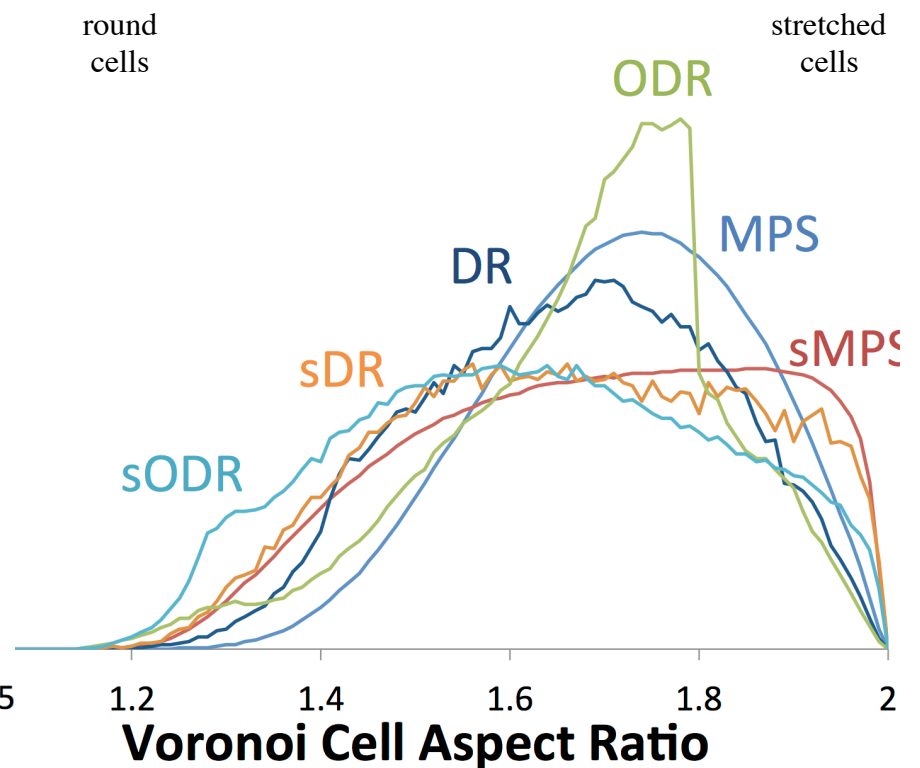


Sifting changes triangulation edge lengths, angles, Voronoi cell squish

shifts Voronoi cell area distribution



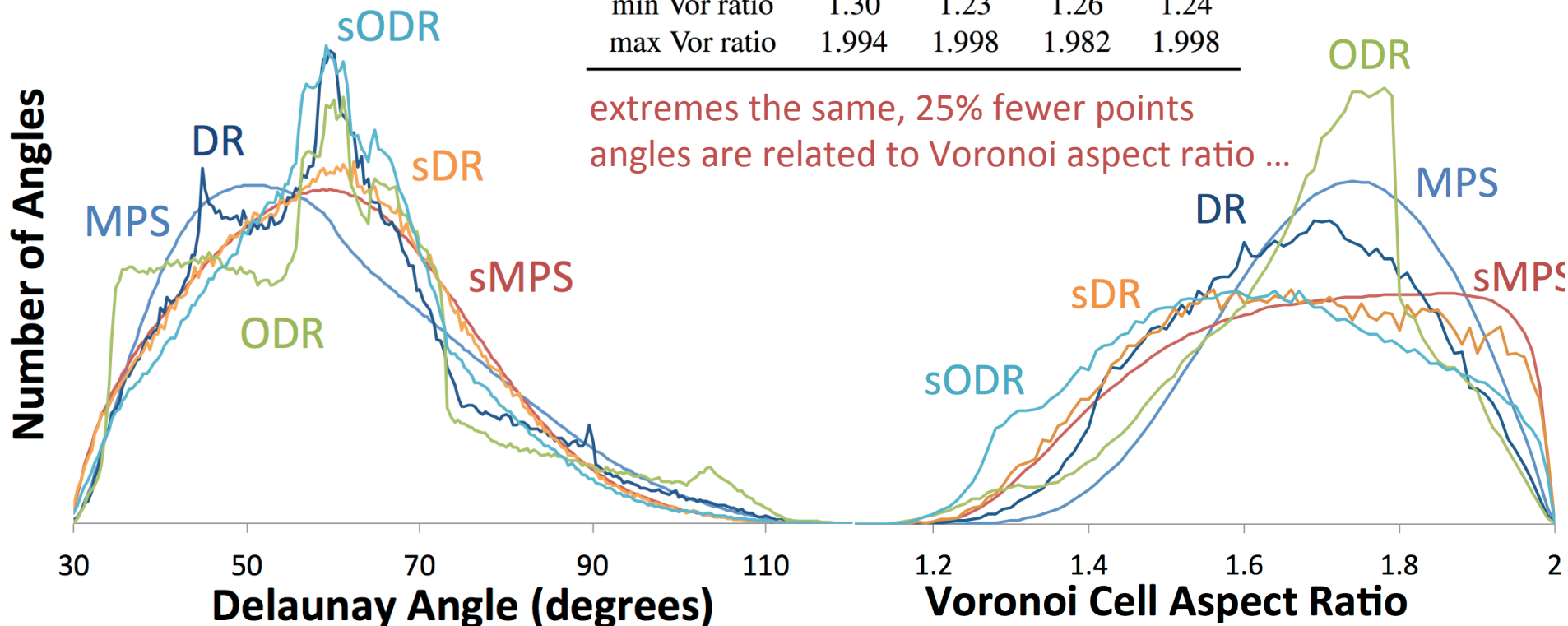
spreads Voronoi cell aspect ratio



Sifting changes triangulation edge lengths, angles, Voronoi cell squish

	MPS	sMPS	DR	sDR
interior points	580	419	580	417
... reduction	-	27%	-	28%
min angle	30.6	30.5	31.7	30.6
max angle	115.5	114.7	110.7	115.5
min Vor ratio	1.30	1.23	1.26	1.24
max Vor ratio	1.994	1.998	1.982	1.998

changes angle distributions





How Well Did We Do?

- **Goal 2: Randomness**

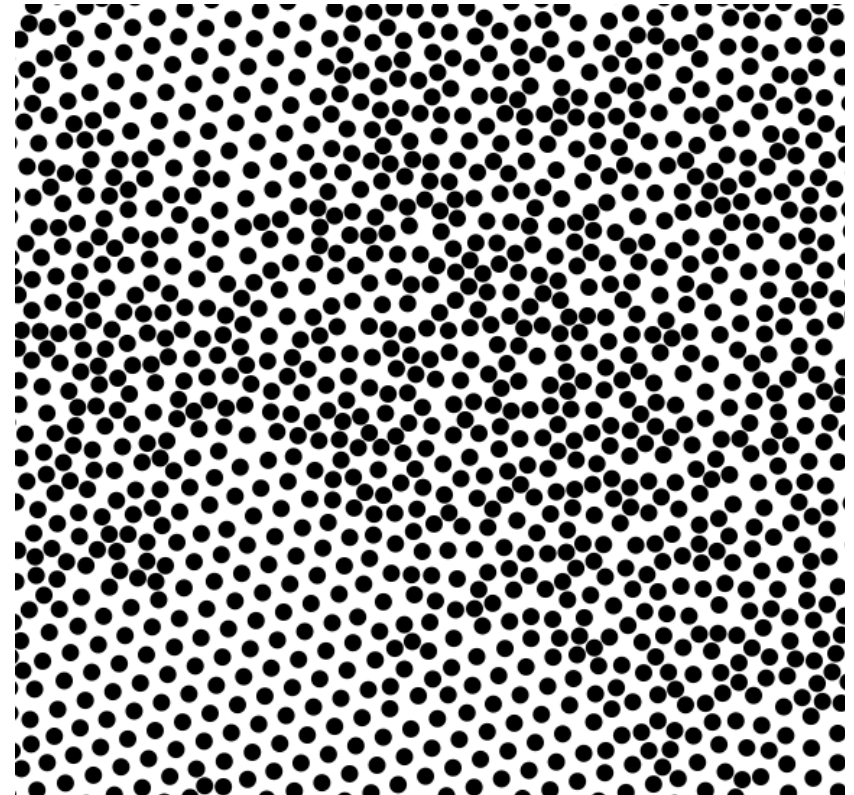
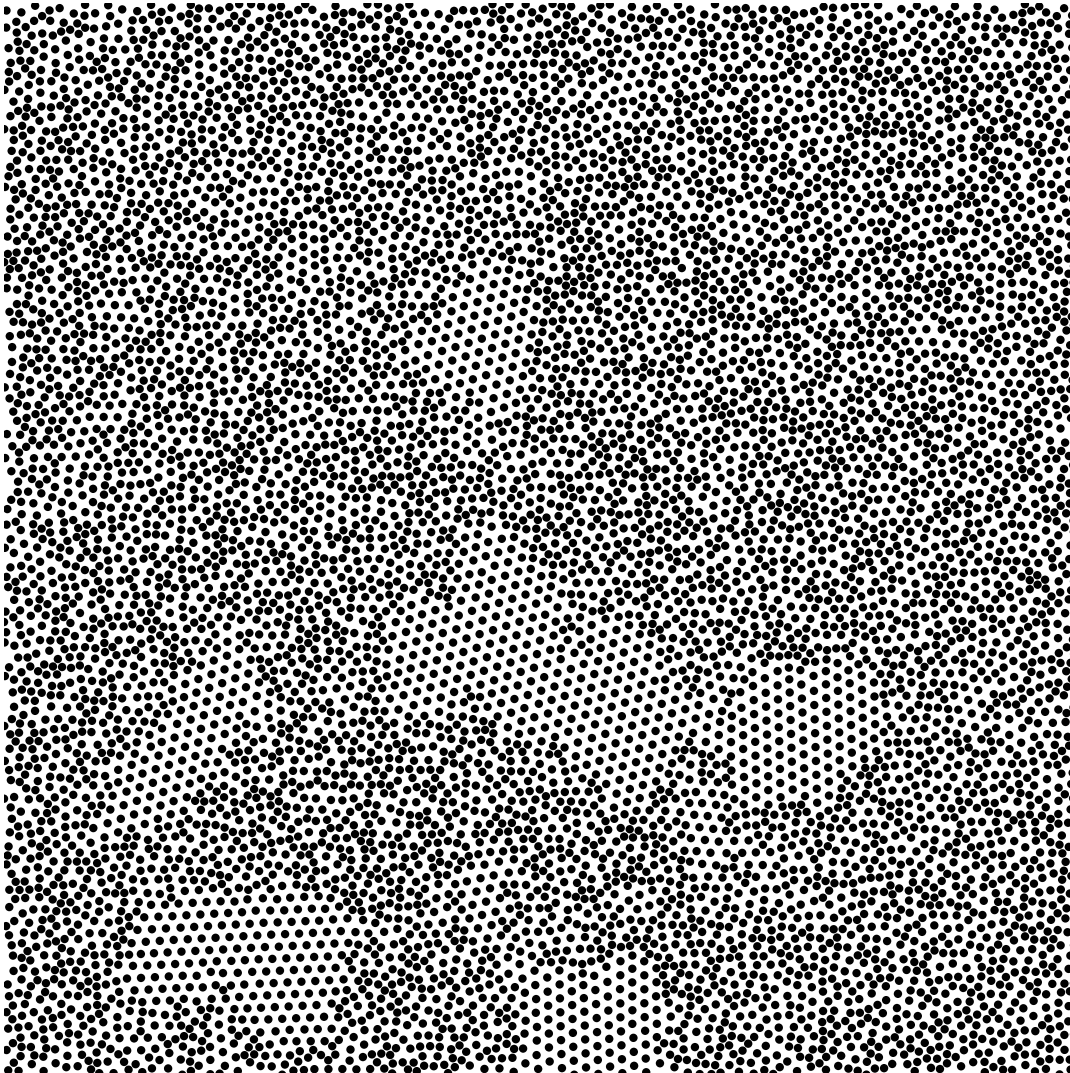


DR and ODR

Sometimes Appear Random

- Many control parameters
 - Which circle (off) center to insert next?

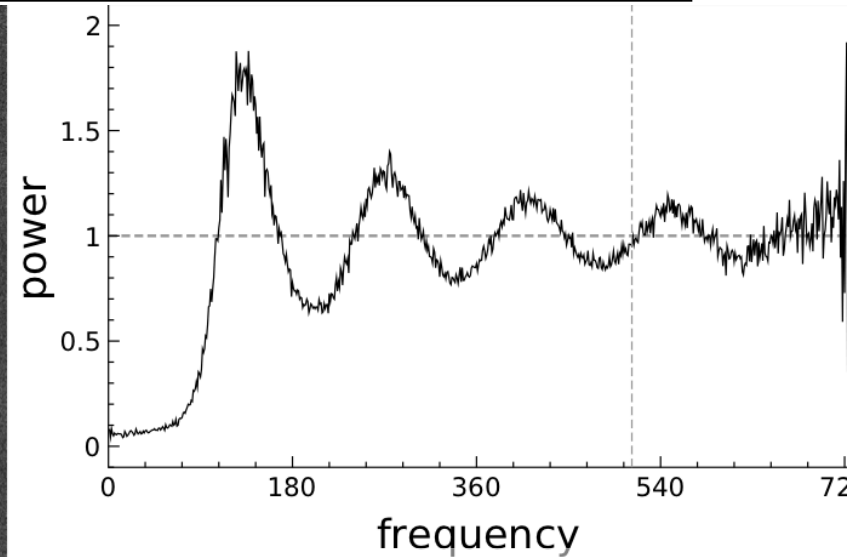
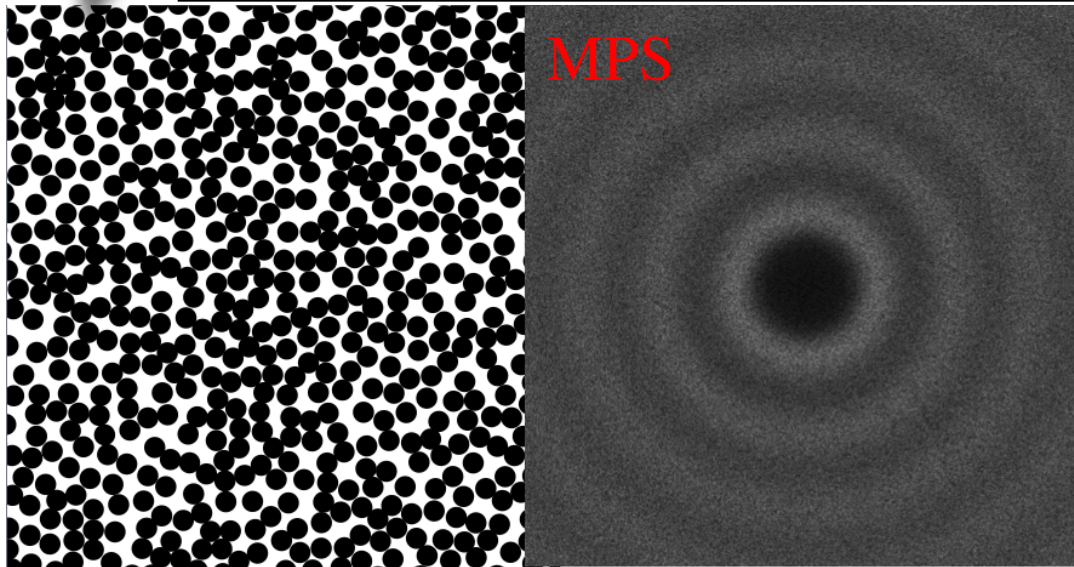
Start with random-looking
DR, ODR
Not these!



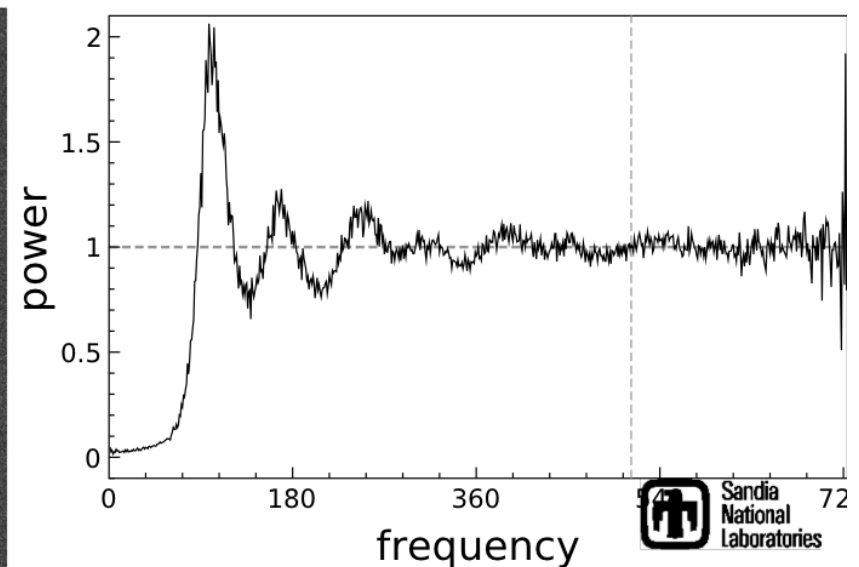
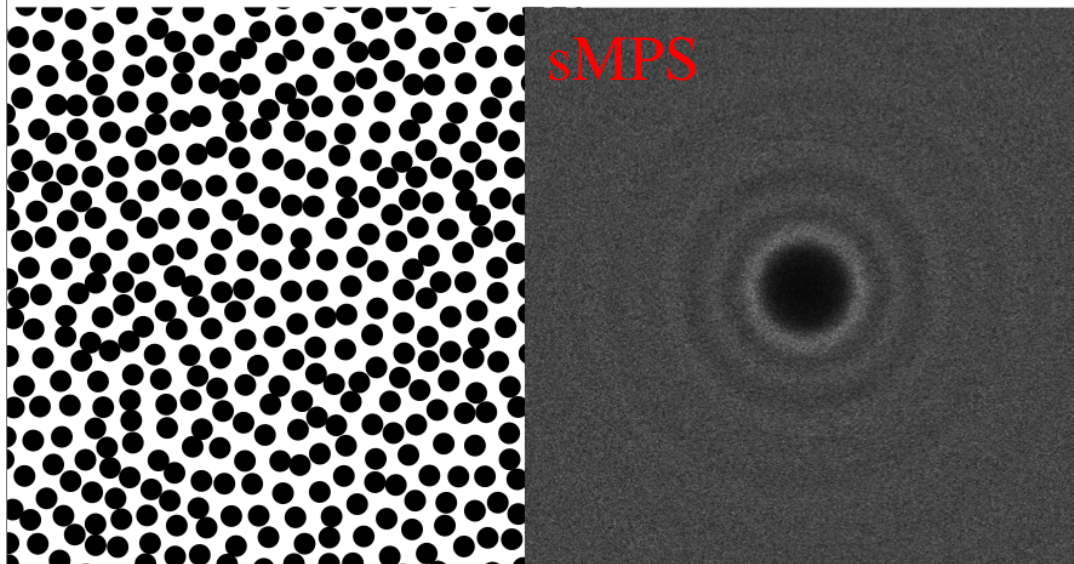
ODR

Sifting Retains Randomness Surprise! But not identical.

MPS

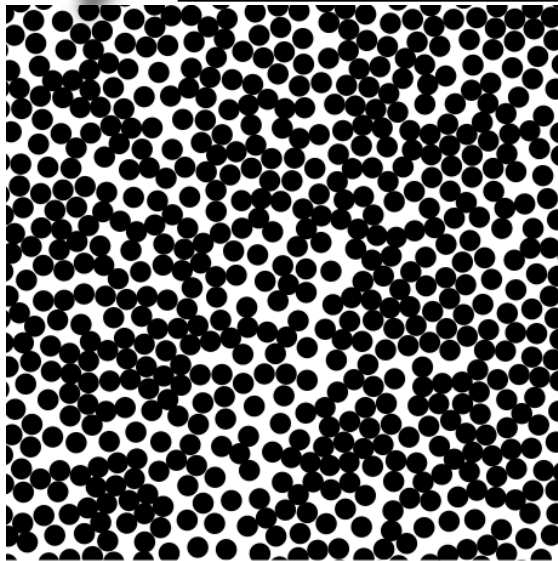


Can you tell me which is "better"?
What's ideal?

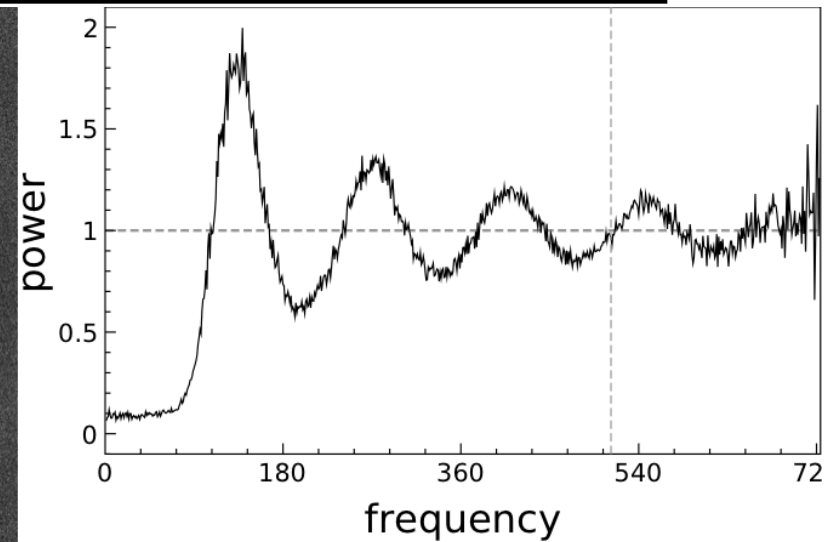
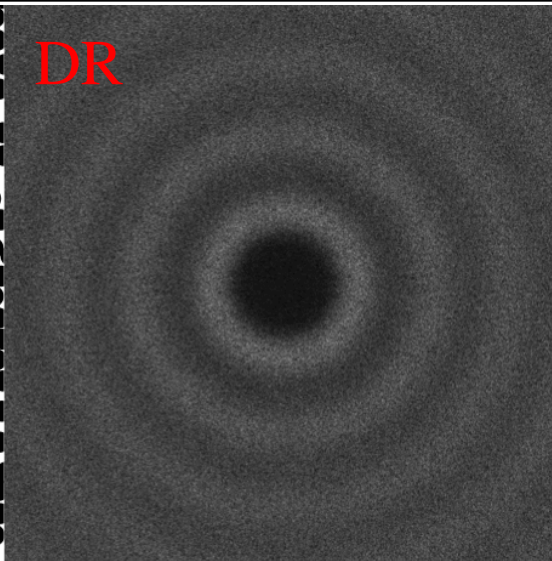


Sifting (introduces?) Randomness Surprise! But not identical.

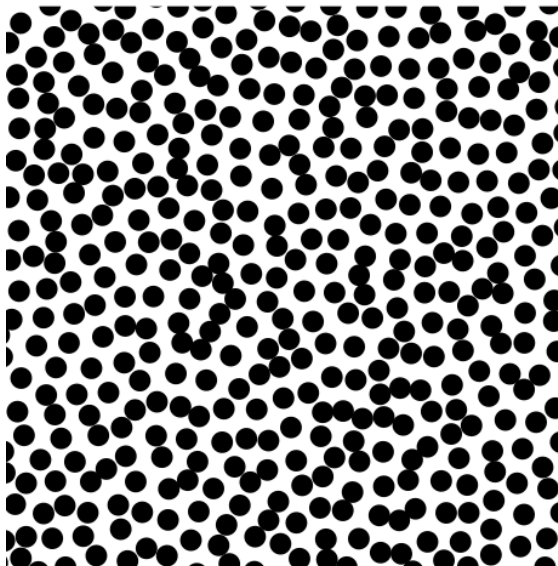
DR



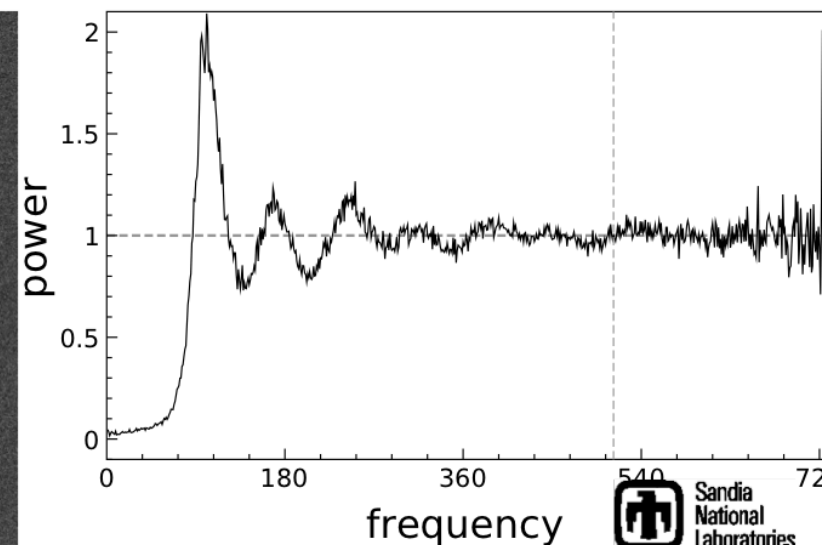
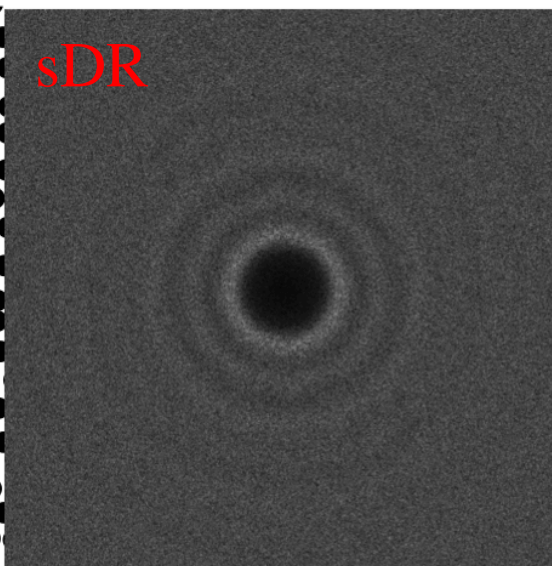
DR



Can you tell me which is “better”?
What’s ideal?

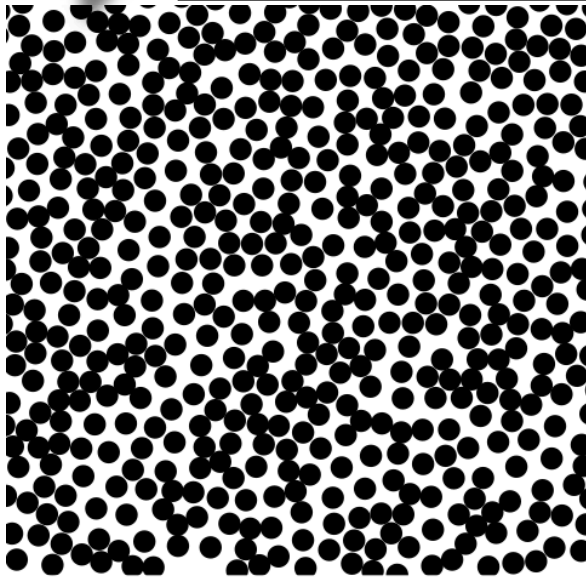


sDR

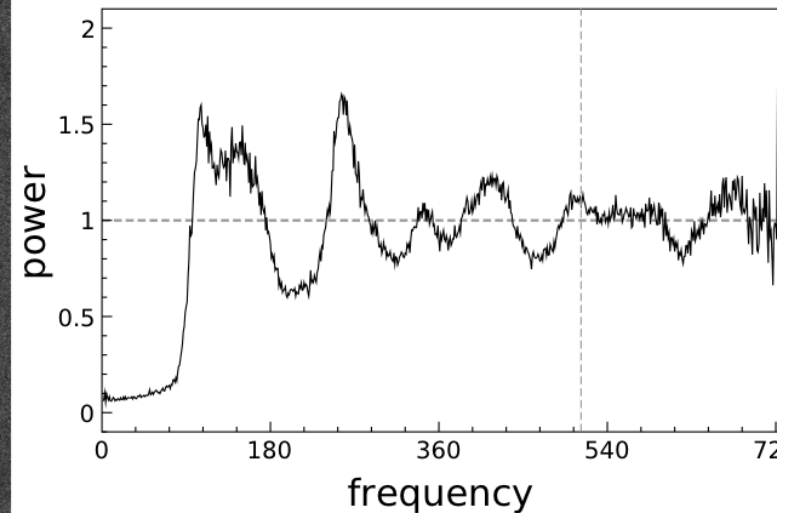
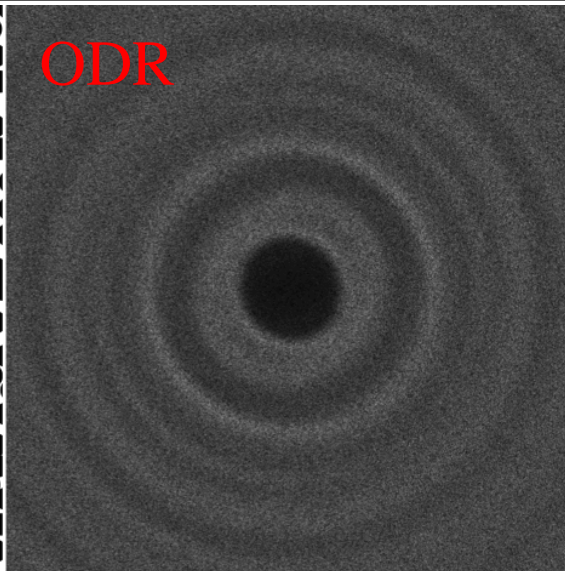


Sifting (introduces?) Randomness Surprise! But not identical.

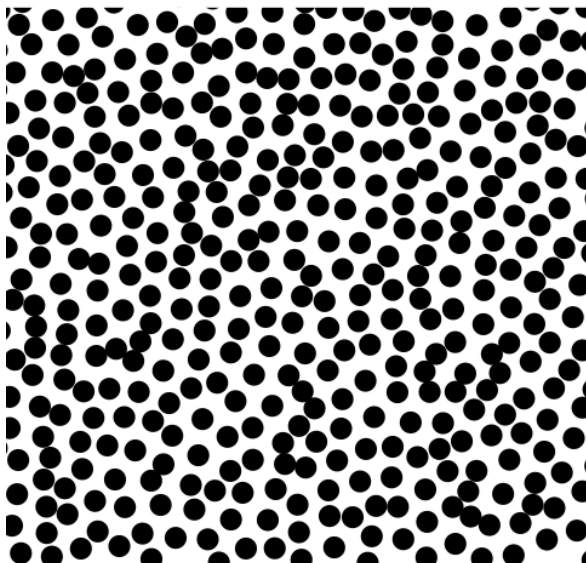
ODR



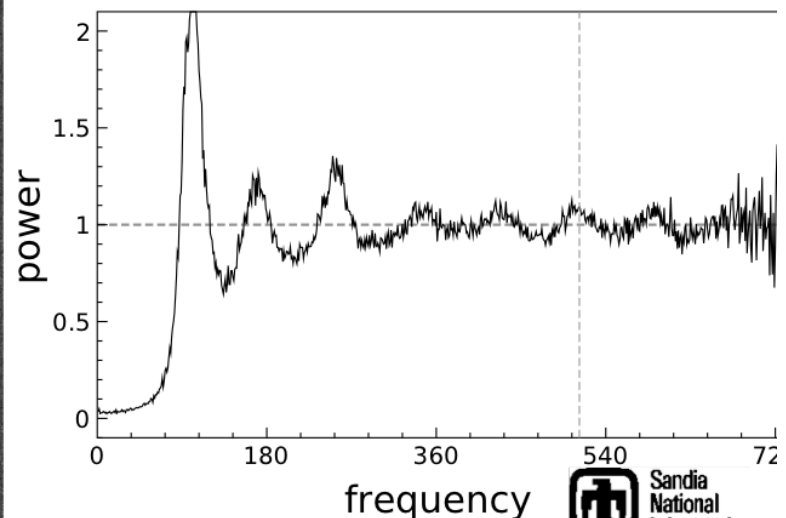
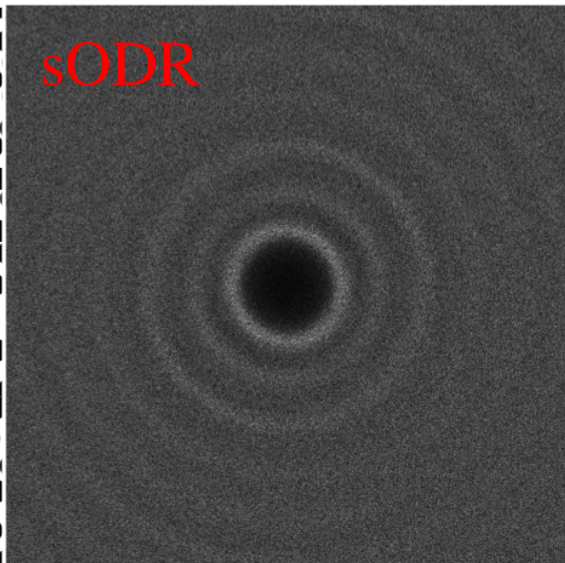
ODR



Can you tell me which is “better”?
What’s ideal?

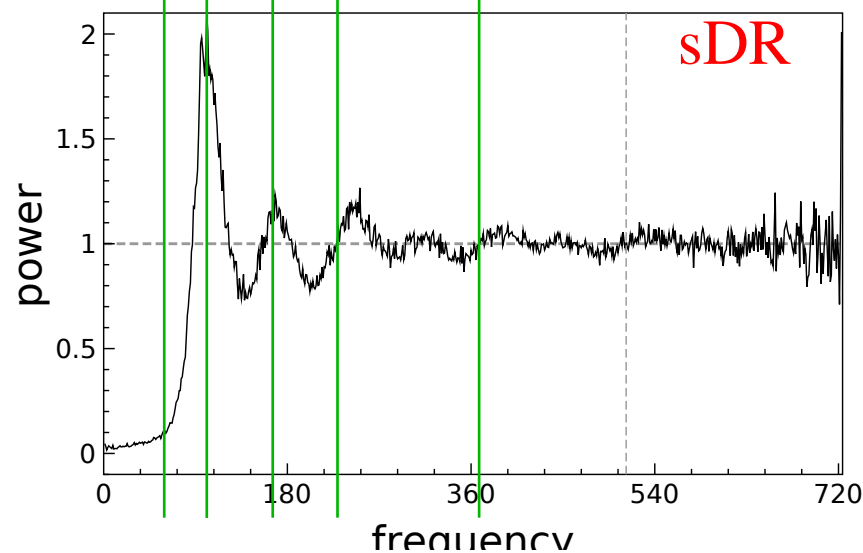
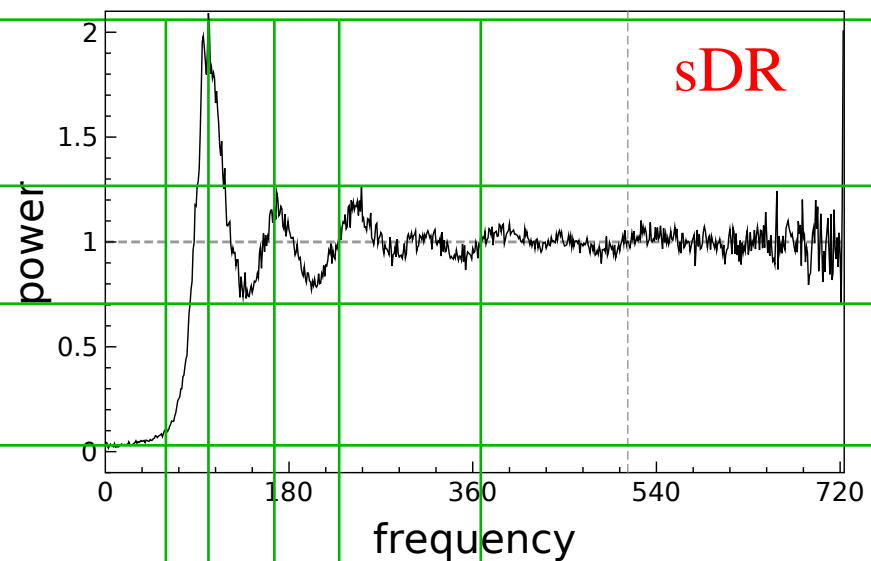
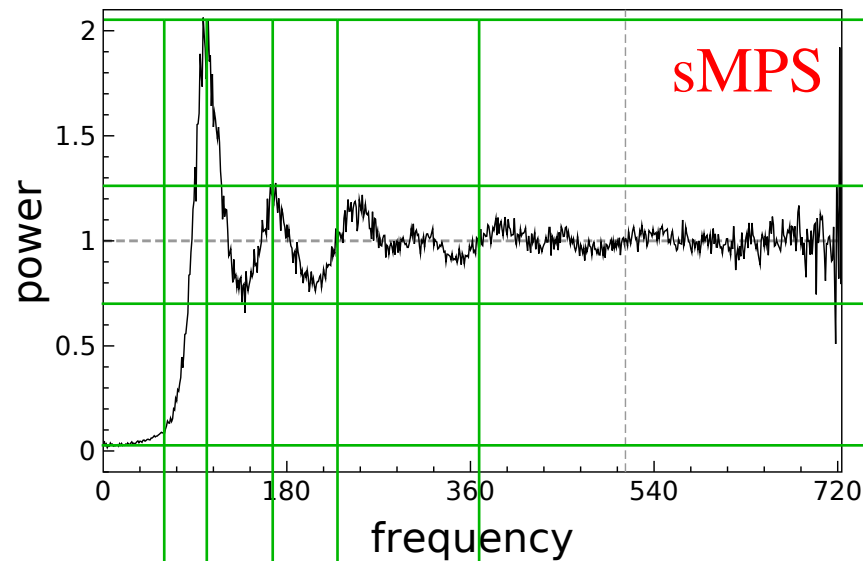


sODR





Original distribution doesn't seem to matter much

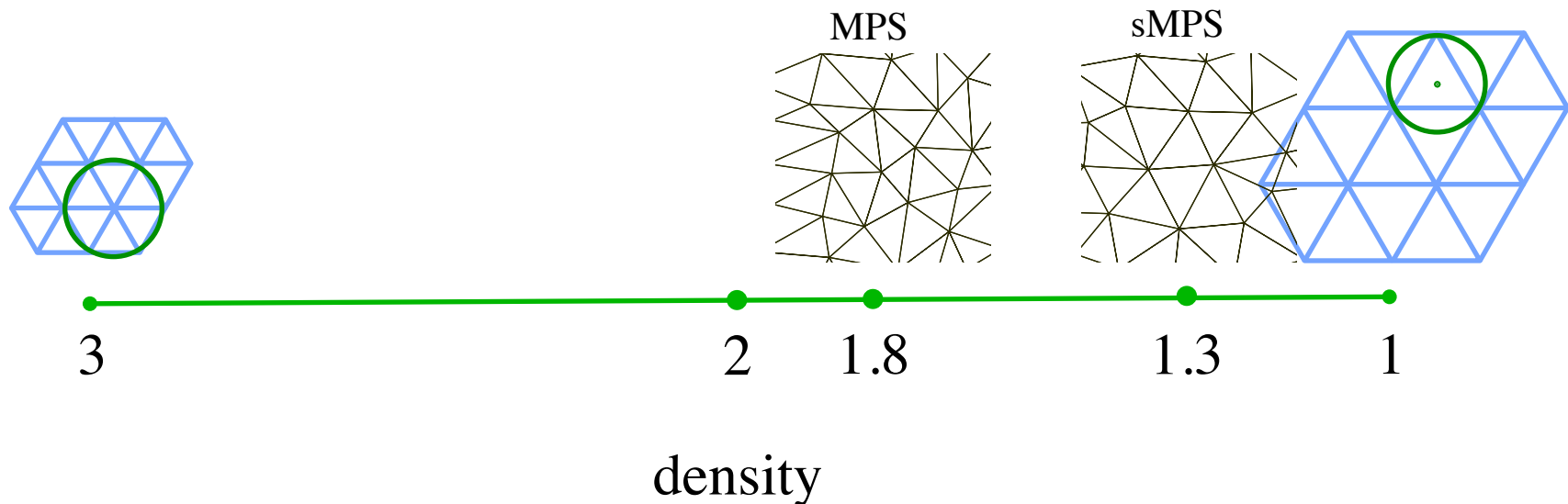


sDR density 1.48
sMPS density 1.33
sODR density 1.33



What's happening?

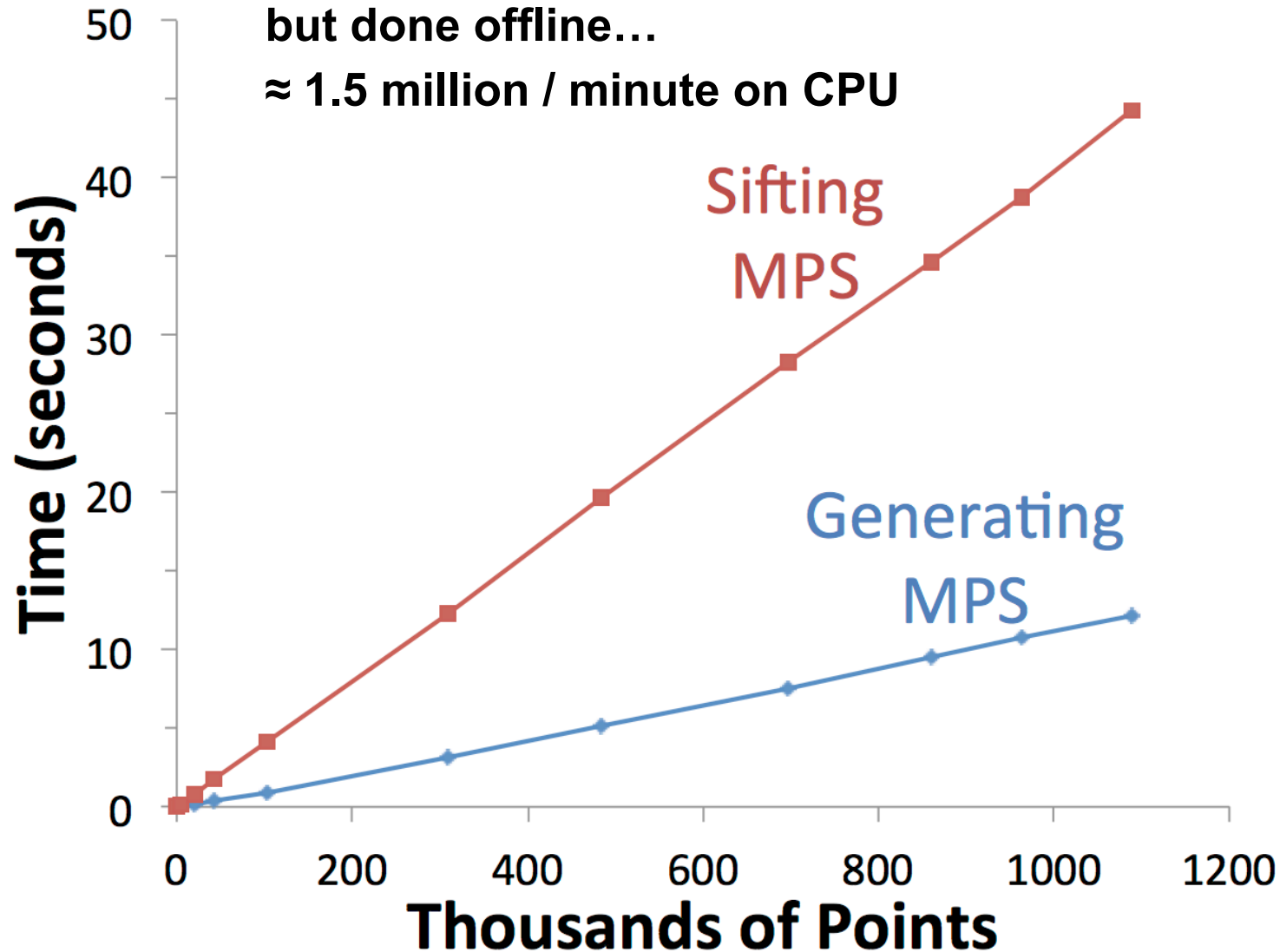
- Gets less dense but never gets close to “converging” to a structured mesh
 - No **pair** can be replaced by **one**.
 - Stuck at a globally random configuration
 - A **triple** can be replaced by **two**? Would we want to?





Time and Memory Effectively Linear

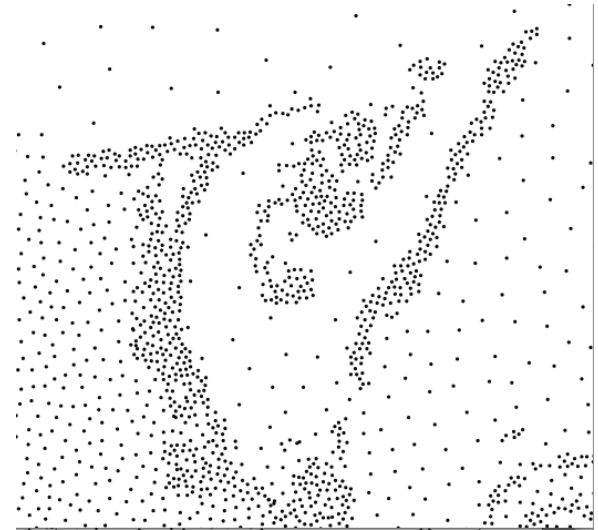
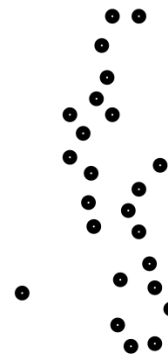
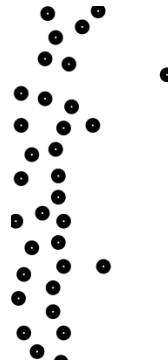
- Sifting 4x slower than generating MPS but done offline...
≈ 1.5 million / minute on CPU



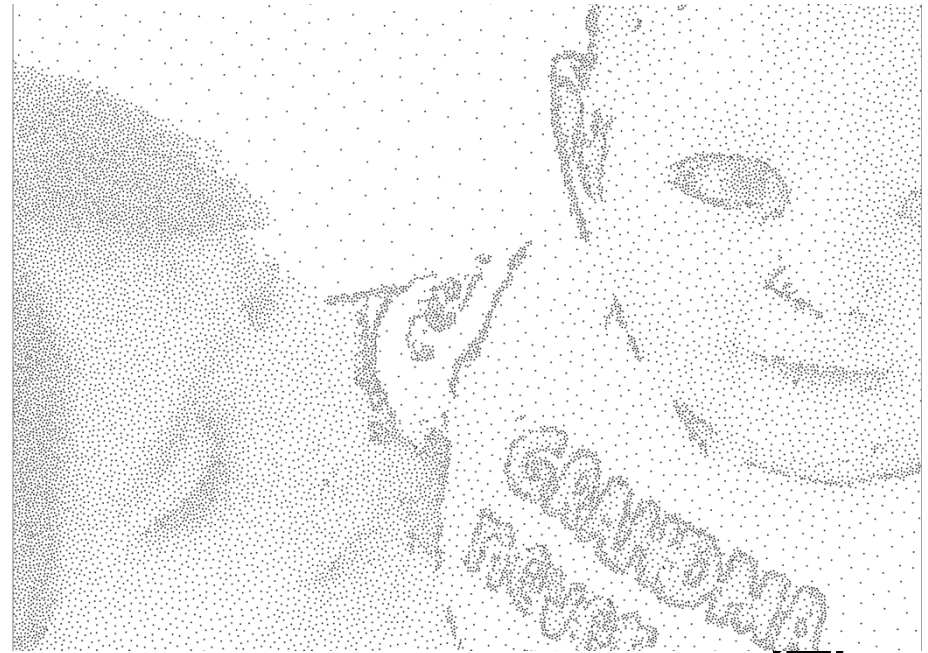


Beyond Uniform

- Prior explanation constant radius
- Spatially varying radii
 - Theory
 - Maximum rate of change L
 - $L = 1, 1/2$
 - Stippling application
 - L exceeds theory limit, still works
 - Disk radius ratio = 10



grayscale sizing function, high contrast

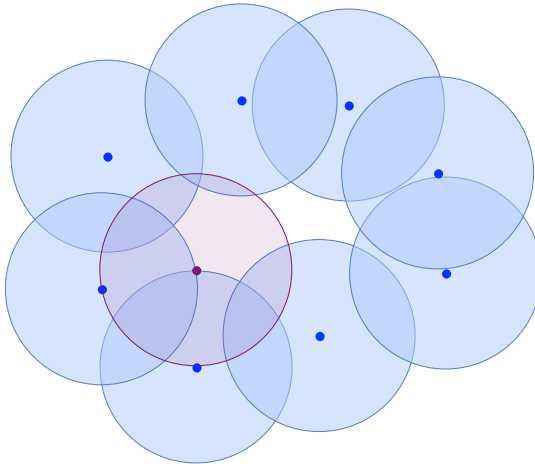


abrupt density changes



Beyond 2d

- Prior was all 2d
 - Higher dimensions
 - Seems straightforward to implement
 - Too many purple points for $d > 5$? 6? Kissing number $\approx 3^d$
 - Effectiveness unknown



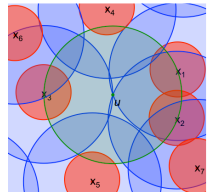


Bonus Thought how I think about sampling

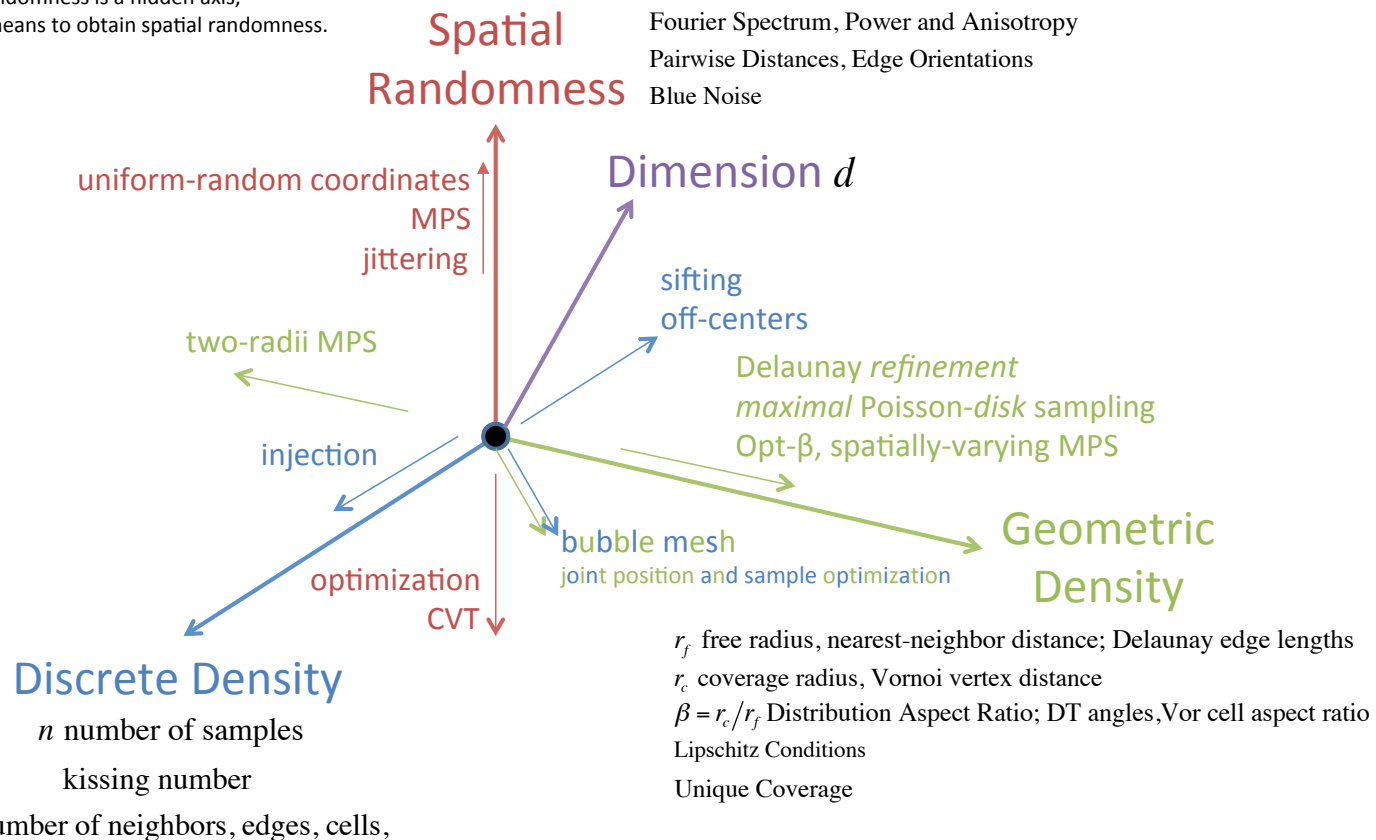
Scott: reorganized this so metrics are in black, the concept being measured in large colored text, and techniques in smaller non-cap text.
One can imagine both local and global measures, min max ave dev, for each of the axes and their metrics, some more natural than others.

A Space for All Sampling Methods

Process randomness is a hidden axis,
merely a means to obtain spatial randomness.



Variable Radii MPS
(two-radii variant)
Mitchell et al.
CCCG 2012

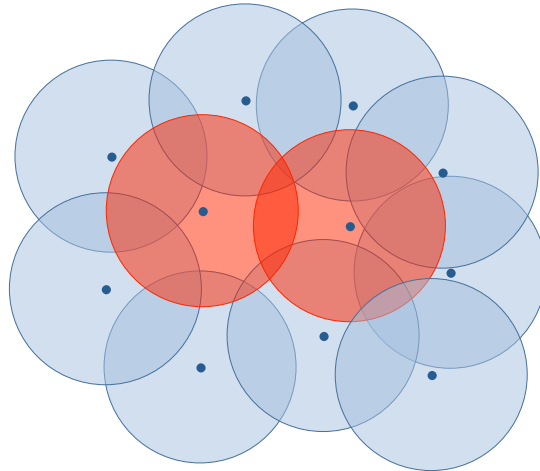




Summary

- **Sifting (replace 2-for-1) points**
 - Reduces the number of points
 - Retains randomness and quality
 - Poisson-disk sampling as a subroutine - resample
- **To do**
 - Theory for rapidly varying sizing function, $L \gg 1$
 - High dimensions
 - Generate a sparser distribution to begin with

- Representative image for website



Sifted Disks

